

Analysis and Simulation of an Encrypted State Cloning Protocol

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1 Abstract

Quantum computing has promising applications for optimizing tasks that would classically require large amounts of computational resources. Reliable transfer of quantum states is integral to large-scale quantum computing. Recently, it has been shown that one can encrypt an unknown wavefunction for cloning and transportation. Decryption of only one of these clones is possible, allowing the creation of multiple copies of a wavefunction without violating the no-cloning theorem. This paper explores the mathematics behind the encrypted state cloning and decryption, as well as a simulation of this process.

2 Introduction

2.1 Quantum Computing

Quantum Computing is a relatively new section of computer and information sciences that leverages the principles of quantum mechanics to achieve higher computational efficiencies over many classical algorithms. The full scope of quantum computing is yet to be fully characterized. Still, advantages in microscopic systems simulations, brute force searching, the Fourier transform, and many more algorithms have already been realized.

Quantum algorithms rely on manipulating qubits, or quantum bits, as opposed to classical bits. In practice, classical bits can be any two-state system. Qubits, however, are required to be objects whose choice of state can be placed in a superposition quantum mechanically. A lot of the power in a quantum algorithm comes from the operations that can be performed on a qubit compared to a classical bit. When performing computations on a classical bit, a bit can only be turned on or off. A qubit's **superposition of state** can be modified such that the superposition is related to a correct output, and this can be achieved with fewer total operations in many cases.

Quantum algorithms are composed of gates, each of which operates on a set of qubits to modify their wavefunctions. The most fundamental of these gates are the X, Y, and Z gates. These are single-qubit gates that are simple rotations in a three-dimensional vector space, and are used to change the computational basis of your system.

2.2 Superposition of States

A superposition of states for a qubit can be expressed as a wavefunction, $|\psi\rangle$. Mathematically, $|\psi\rangle$ can be expressed as several objects. The first is a linear combination of the states that comprise the superposition, or **wavefunction**:

$$|\psi\rangle = \sum c_i |x_i\rangle$$

where c_i is the probability amplitude corresponding to the state $|x_i\rangle$. In quantum computing, qubits only occupy one of two states for each of the three

measurement bases. In the Z basis, these states are the $|0\rangle$ state ("on" state), or the $|1\rangle$ state ("off" state). For example, in the Z basis, one qubit has the wavefunction.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

The information in the wavefunction can be further condensed into a column vector:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

and the entries of this column vector are the wavefunction's probability amplitudes. It is important to note that for quantum computing, for a state to be a wavefunction to be a valid quantum state, it must be **normalized**, or:

$$|\alpha|^2 + |\beta|^2 = 1$$

Where $|x|^2 = x \times x^*$ is the norm squared of x , and x^* is the complex conjugate of x . This statement means that the total probability of measuring a state is 1. Another important definition is the conjugate of a state vector, denoted $|\psi\rangle^* = \langle\psi|$:

$$\langle\psi| = [\alpha^* \quad \beta^*]$$

The conjugate of a state vector is actually a row vector. It is also the case that quantum gates can be represented mathematically by matrices.

2.3 Quantum Gates

Quantum gates in quantum computing are a subset of the group of quantum operators. All quantum gates are unitary operators whose job is to manipulate the state of the qubit(s) they act on. An n -qubit gate is represented by a $2^n \times 2^n$ unitary matrix. There are three fundamental one-qubit gates, the X, Y, and Z gates, that are defined as:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

All quantum gates must satisfy the constraint that they are unitary operations represented by unitary matrices. For a matrix, U , to be unitary, it must obey the following rule:

$$UU^\dagger = U[U^{*T}] = I$$

Or, simply put, the hermitian conjugate of a unitary matrix is its own inverse. This implies an important quantum computing concept: that any operation done in a quantum algorithm can be undone.

2.4 The Bloch Sphere

It can be hard to conceptually see what a quantum gate does to the state vector of a qubit. Because, in quantum computing, we use a two-state system to define a qubit:

$$|\psi_{qubit}\rangle = \alpha |0\rangle + \beta |1\rangle$$

We can rewrite α and β such that our state vector transforms into

$$|\psi_{qubit}\rangle = \cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} e^{i\phi} |1\rangle$$

This form maintains normalization and allows us to visualize our qubit's state vector as a point on a three-dimensional unit sphere. This sphere is called the "Bloch Sphere" and is useful for visualizing the evolution of a quantum state throughout an algorithm. The Bloch sphere looks like

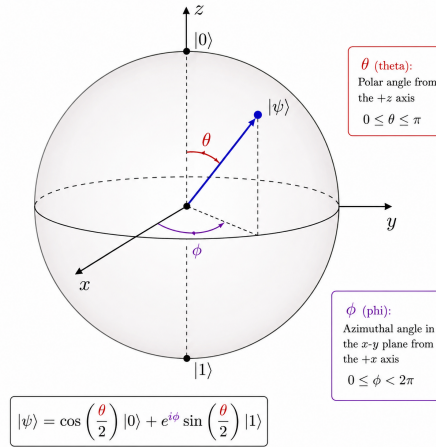


Figure 1: A diagram of the Bloch Sphere with θ and ϕ labeled. With phase errors, these angles can begin to wander or flip entirely.

Where the three axes correspond to a computation measurement basis. And we see the three simplest quantum gates: the X, Y, and Z gates, are simply rotations about their corresponding axes.

2.5 Entanglement

Entanglement is a crucial component of any quantum algorithm. Without it, any quantum circuit would simply be a classical algorithm with incredibly sensitive computational bits. Entanglement is a phenomenon in which the measurement outcome of one qubit is correlated (or anticorrelated) with the measurement of another qubit. Mathematically, we can express a two-qubit system as:

$$|\psi\rangle = \alpha |00\rangle + \beta |01\rangle + \gamma |10\rangle + \delta |11\rangle$$

If the two systems are not entangled, we can express this system as the product of two smaller systems:

$$|\psi\rangle = |a\rangle |b\rangle = (a_1 |0\rangle + a_2 |1\rangle)(b_1 |0\rangle + b_2 |1\rangle)$$

Where the F.O.I.L.-ing of this expression will allow you to retrieve the probability amplitudes α , β , γ , & δ . In the case of an entangled state, you cannot find values for a_1, a_2, b_1 , and b_2 such that

$$\begin{aligned} a_1 b_1 &= \alpha, & a_1 b_2 &= \beta \\ a_2 b_1 &= \gamma, & a_2 b_2 &= \delta \end{aligned}$$

This is a peculiar result: you have a system of equations with no solution, yet it physically exists, and it is the basis of quantum computations.

2.6 Bell States

In quantum computing, four maximally entangled states are integral to some quantum computing algorithms. These states are the $|\Psi^+\rangle$, $|\Psi^-\rangle$, $|\Phi^+\rangle$, and $|\Phi^-\rangle$ states. These states consist of two qubits and are expressed as

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\ |\Phi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\ |\Psi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \end{aligned}$$

In the $|\Psi^\pm\rangle$ states, the measured outcome of the two qubits is correlated, meaning the second qubit will always collapse to the same state as the first qubit, and in the $|\Phi^\pm\rangle$ state, the qubits are anti-correlated. The $|\Psi^+\rangle$ is integral to quantum state teleportation.

2.7 Tensor Products

Oftentimes, it is easy to describe a combination of gates as a tensor product of the two individual gates. Denoted by $\otimes$, the tensor product is a way to combine

vector spaces. Given two Matrices (of not necessarily even dimensions), A and B, their tensor product $A \otimes B$ is:

$$A \otimes B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \otimes \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a \begin{bmatrix} e & f \\ g & h \end{bmatrix} & b \begin{bmatrix} e & f \\ g & h \end{bmatrix} \\ c \begin{bmatrix} e & f \\ g & h \end{bmatrix} & d \begin{bmatrix} e & f \\ g & h \end{bmatrix} \end{bmatrix}$$

2.8 Matrix Exponentiation

It will become relevant during our encryption scheme to define the exponentiation of a matrix. Let X be a unitary matrix (as X must be unitary for it to be a valid quantum gate), then its exponential e^X can be defined by its Taylor expansion:

$$e^X = \sum_{n=0}^{\infty} \frac{X^n}{n!}$$

Because X is a unitary matrix, it is also the case that it is diagonalizable. Expanding $X = PDP^{-1}$:

$$\begin{aligned} e^X &= \sum_{n=0}^{\infty} \frac{(PDP^{-1})^n}{n!} \\ &= \sum_{n=0}^{\infty} \frac{PD^nP^{-1}}{n!} \\ &= Pe^D P^{-1} \end{aligned}$$

where D is a diagonal matrix consisting of the eigenvalues of X:

$$D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda_N \end{bmatrix}$$

2.9 Measurement

Measurement has been mentioned thus far without defining what it means to measure a quantum system. Consider a general wavefunction

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Then when we measure $|\psi\rangle$, we have two possible possible outcomes: $|0\rangle$ with probability $|\alpha|^2$ or $|1\rangle$ with probability $|\beta|^2$. Measurement of this system also collapses our superposition state to a classical state. This simply means that, if, for example, we measure the system in the $|0\rangle$ state, any subsequent

measurements will always yield the same result. To state measurement plainly, measurement is the process of collapsing your wavefunction to one of the possible states that were superposed.

3 The No-Cloning Theorem

In quantum mechanics, there is an important theorem known as the no-cloning theorem. This theorem states that A quantum state from one system can not be cloned identically in another system without disrupting the original wavefunction. Consider a general wavefunction $|\psi\rangle$, and another system, $|\psi_{clone}\rangle$. To extract quantum information from $|\psi\rangle$, you will need to perform an operation U on both systems:

$$U |\psi\rangle \otimes |\psi_{clone}\rangle$$

That preserves all quantum amplitudes of $|\psi\rangle$ while transporting information to $|\psi_{clone}\rangle$. This is not possible, as any operation that interacts with both systems would induce some change in both systems. Therefore, you cannot clone a quantum state, but can "teleport" a state between two qubits.

As a short proof, let us assume that there is a Unitary operator, U, that can clone a state $|\psi\rangle$. For U to be a valid quantum operator, it must be able to clone different states (U must be a linear operator) $|\phi\rangle$, or:

$$U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle$$

and

$$U |\phi\rangle |0\rangle = |\phi\rangle |\phi\rangle$$

We can now take the inner product of both sides of these two relations, which must be equivalent:

$$(\langle\psi| \langle 0| U^\dagger)(U |\phi\rangle |0\rangle) = \langle\psi| \langle\psi| |\phi\rangle |\phi\rangle$$

This now becomes

$$\langle\psi|\phi\rangle = (\langle\psi|\phi\rangle)^2$$

we see that $|\psi\rangle$ must be either equivalent or completely orthogonal to $|\phi\rangle$. This, however, cannot occur for an arbitrary state.

4 Quantum State Teleportation

Suppose that a person, named Alice, would like to send another person, named Bob, a qubit in a quantum state that she has prepared in advance. Since this process works for an arbitrary state, Alice may not know her qubit's state, so she cannot tell Bob how to prepare any of his own qubits to replicate Alice's qubit. Alice can instead transport her qubit's arbitrary state, $|\psi\rangle$, to one of Bob's qubits

via quantum state teleportation. Broadly, quantum state teleportation relies on the entanglement of Alice's qubit with a Bell state, which is well characterized, to allow Bob to easily take the state of Alice's qubit and "move" it to one of his own. It is very important that Alice's qubit is **not** being copied. This would violate the no-cloning theorem.

The process of quantum state teleportation starts with three qubits. Alice's qubit is in the state, $|\psi_A\rangle = \alpha|0\rangle + \beta|1\rangle$, while the other two qubits are in the Bell state $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. The total state of the system is thus

$$\begin{aligned} |\psi\rangle &= (\alpha|0\rangle + \beta|1\rangle)\left(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\right) \\ &= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle \end{aligned}$$

Now, Bob can take the third qubit, belonging to the entangled pair, leaving Alice with the first two qubits. Alice can then apply a C-NOT gate (Controlled NOT gate) to her two qubits, changing the state of all three qubits to:

$$\begin{aligned} |\psi\rangle &= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|110\rangle + \frac{\beta}{\sqrt{2}}|101\rangle \\ &= \frac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle) \\ &= \frac{1}{\sqrt{2}}(\alpha(|000\rangle + |011\rangle) + \beta(|110\rangle + |101\rangle)) \end{aligned}$$

Alice now applies a Hadamard gate to her first qubit, which is a rotation on the Bloch sphere about the axis $\frac{1}{\sqrt{2}}(\hat{x} + \hat{z})$ and changes the state to:

$$\begin{aligned} H|\psi\rangle &= \frac{1}{\sqrt{2}}[\alpha(|+00\rangle + |+11\rangle) + \beta(|-10\rangle + |-01\rangle)] \\ &= \frac{1}{\sqrt{2}}[\alpha\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)(|10\rangle + |01\rangle)] \\ &= \frac{1}{2}[\alpha(|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta(|0\rangle - |1\rangle)(|10\rangle + |01\rangle)] \\ &= \frac{1}{2}[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\beta|0\rangle + \alpha|1\rangle) \\ &\quad + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(-\beta|0\rangle + \alpha|1\rangle)] \end{aligned}$$

This final representation of our system may feel intimidating, but in reality, it conveys a simple idea. If we only measure the first two qubits in our system (Alice's qubits), there are only four possible outcomes. Alice's qubits can be in the states $|00\rangle$, $|01\rangle$, $|10\rangle$, or $|11\rangle$, All with probability $|\frac{1}{2}|^2 = \frac{1}{4}$. This fact creates a case-by-case decision tree to place the last qubit (Bob's qubit) in the original state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

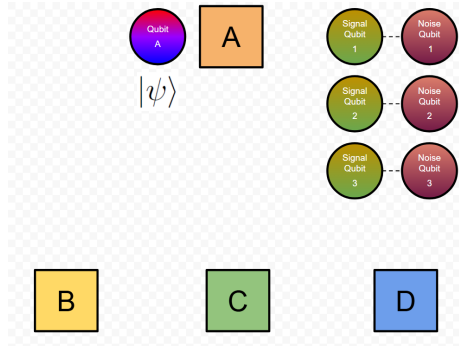
Measured State of the first two qubits	Operation to teleport $ \psi\rangle$
$ 00\rangle$	I-Gate (Nothing)
$ 01\rangle$	X-Gate
$ 10\rangle$	Z-Gate
$ 11\rangle$	X-Gate then Z-Gate

This decision tree can be implemented by adding a controlled X-gate and a controlled Z-gate at the end of our quantum algorithm. The completed quantum algorithm looks as follows:

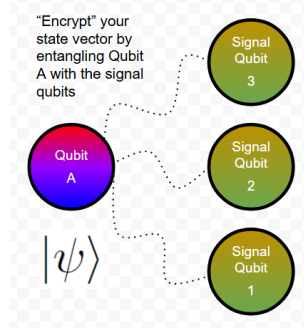
5 Encrypted State Cloning

Here, we will describe the process of cloning encrypted states. This procedure is highly analogous to quantum state teleportation, and roughly follows these steps:

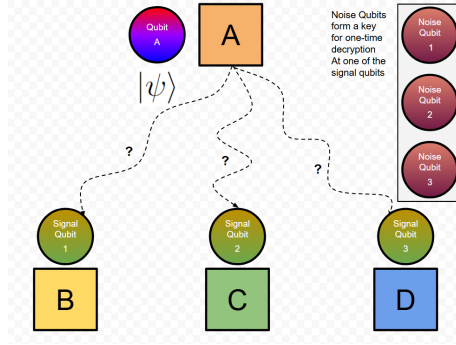
1. Entangle signal qubits (that carry information on your original state) with a Bell state



2. Distribute your signal qubits however you desire for future decryption



3. Decrypt one of your signal qubits to retrieve the original wavefunction $|\psi\rangle$



The only difference between this process and quantum state teleportation, as described in the previous section, is that multiple qubits (similar to Bob's qubit) are created to create options about where Alice can send her original wavefunction. This process looks as follows:

Quantum information is imprinted on the set of signal qubits, and only the signal qubits. However, because of the entanglement that exists between the signal and noise qubits, all noise qubits can be used to "decrypt" one of the signal qubits, retrieving Alice's wavefunction, $|\psi\rangle$.

5.1 Encrypted States

Recall that the beginning of the quantum state teleportation process requires that Alice's qubit be entangled with the Bell state $|\Phi^+\rangle$. To encrypt a state, you first begin by preparing n pairs of qubits entangled in the $|\Phi^+\rangle$ state. One-half of these pairs of qubits are referred to as "signal qubits" (denoted by S_i), and the other half are considered "noise qubits" (N_i). The purpose of the signal qubits is to have quantum information about Alice's wavefunction $|\psi\rangle$ imprinted

on them. The noise qubits then form a key to retrieve $|\psi\rangle$ from one, and only one, of the signal qubits. To imprint the $|\psi\rangle$ on your signal qubits, you perform the following encryption operation:

$$U_{encrypt} = e^{-\frac{\pi}{4}\sigma_1^{(A)} \otimes (\bigotimes_{i=1}^n \sigma_1^{(S_i)})} e^{-\frac{\pi}{4}\sigma_3^{(A)} \otimes (\bigotimes_{i=1}^n \sigma_3^{(S_i)})}$$

where σ_1^W is the X gate and σ_3^W is the Z gate applied to the W'th qubit. Note that these matrix exponentials form operations that are only applied to Alice's qubit, A, and the signal qubits, S_i . The state of the noise qubits remains unchanged. This state of your system is considered to be the encrypted state of Alice's qubit and the signal qubits.

After Encryption, all signal qubits are left in a mixed state. If they were, in fact, left in a pure state, then they would be decryptable without the use of the noise qubits as a key.

5.1.1 Density Matrices

The density matrix contains valuable information about a quantum state. Consider a general state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then the probability density matrix, $\rho(|\psi\rangle)$ is defined:

$$\rho(|\psi\rangle) = |\psi\rangle \langle\psi| \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}$$

because our main diagonal is simply the probability amplitudes of $|\psi\rangle$, it must be true that the trace, $Tr(\rho) = 1$. The probability density matrix is a component that is unique to a pure quantum state and can be used to identify the fidelity of a measured result with an expected outcome.

5.1.2 Mixed States

Recall the image of the Bloch Sphere

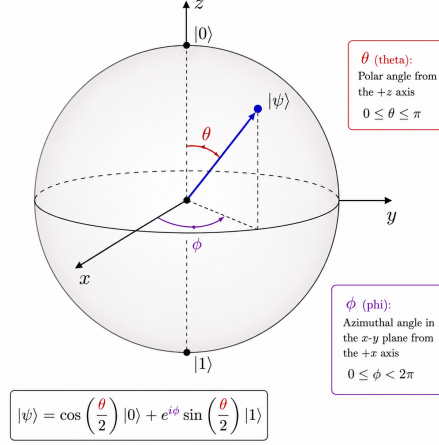


Figure 2: A diagram of the Bloch Sphere with θ and ϕ labeled. With phase errors, these angles can begin to wander or flip entirely.

A quantum state can be thought of as a point on this sphere. There is a way to distinguish between two types of states: mixed and pure states. A pure state on the Bloch Sphere is represented by a point on the surface of the Bloch Sphere. A mixed state is considered a point inside the Bloch Sphere, when some quantum information is missing or lost. In our case with encrypted state cloning, the signal qubits are left in a mixed state because they are entangled with the noise qubits (recall they were prepared in a Bell state), and they comprise a subsystem of the whole set of Alice's qubit, themselves, and the noise qubits.

5.2 Decrypting a Signal Qubit

After Alice has decided where she would like to move her wavefunction, she then uses all noise qubits and the designated signal qubit to perform the following operation:

$$U_{\text{decrypt}}^{(n)} := \sum_{\mu=0}^3 \alpha_{\mu} (|\phi_{\mu}\rangle\langle\phi_{\mu}|_{S_1 N_1}) \otimes \left(\bigotimes_{j=2}^n \sigma_{\mu}^{(N_j)^{\top}} \right)$$

There are several terms in this decryption operator that we must define as well:

$$\alpha_0 = 1, \quad \alpha_1 = \alpha_3 = i, \quad \alpha_2 = (-i)^{n+1}$$

$$|\phi_\mu\rangle_{S_i N_i} := \sigma_\mu^{(S_i)} \otimes I^{(N_i)} |\phi\rangle_{S_i N_i}$$

After performing this operation on the n 'th Signal qubit, Alice's original wavefunction will have been teleported to the qubit S_N . The qubits that formed the decryption key can no longer be used to teleport $|\psi\rangle$ without re-encryption, as the information imprinted on the signal qubits has been changed. This ensures that the no-cloning theorem is not violated.

5.3 Simulation of Encrypted State Cloning

For this project, we have also created a simulation of this algorithm to show that its implementation is feasible. To do so, we utilized Qiskit, a software development kit (SDK) developed by IBM. The primary function of Qiskit is to interact with quantum computers, allowing programming and execution of circuits on real quantum computers. We have managed to successfully encrypt the state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, and retrieve it from the first signal qubit with up to ten noise and signal qubits.

Further work could include implementing this for a signal qubit of the user's choice. This would allow for the use of this program to implement the full encrypted cloning protocol on a quantum computer when the fault tolerance of real-world quantum computers improves.

6 Real World Application of Encrypted State Cloning

Even though the process of encrypting a wavefunction in a set of qubits, and subsequently being able to choose which qubit the original wavefunction is sent to, is interesting, there are questions about the practicality of implementing this protocol. Firstly, the problem being solved by the simplest case of quantum state teleportation is that of moving a qubit (in physical space) can disturb its quantum wavefunction, and thus you should avoid moving the qubit with important information stored within it. With this state-teleportation procedure, you must move all signal qubits. The more signal qubits you have in your system, the more likely it is that some error will occur in the information stored in the wavefunction of your signal qubits. Another problem is that to perform the decryption operation, you need to perform operations on the noise qubits as well as the signal qubit that you are decrypting. This means you must also physically relocate all noise qubits. Once again, as you have more noise qubits, the probability of retrieving the initial quantum state with high fidelity decreases.

This procedure may seem impractical in the current era and near future of quantum computing. This may not be the case forever, however. After the Noisy-Intermediate-Scale-Quantum (NISQ) era comes to a close, it may be

practical to move qubits within a physical system over relatively long distances. This may also hold if photons are realized as effective qubits in practice, which could be more resilient to noise during travel.

Despite these two competing viewpoints, the protocol itself does have some potential applications. It may be possible to have a quantum algorithm that would rely on conditional decryption of a quantum state, where, in the implementation of a physical qubit, moving the noise qubits would be more efficient and useful than moving Alice’s qubit to where it is needed. Additionally, in a future where fault-tolerant quantum computers exist, it could suffice to have geographically differing cloud storage systems, where you need the wavefunction of Alice’s qubit to access information on the cloud, or perform operations [3]

7 conclusion

The encrypted state cloning process has the potential to be a useful procedure for conditional quantum state teleportation. The simulation presented in this paper confirms the theoretical predictions of Yamaguchi and Kemp for up to $n = 10$ Bell pairs, achieving complete retrieval of the original wavefunction. Future work could include testing of this algorithm with different original wavefunctions, testing the procedure on real quantum computers for low numbers of Bell pairs, and investigating the fidelity scaling of the decrypted state under different noise models.

References

- [1] Michael Nielsen and Isaac Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 2010.
- [2] Thomas Wong *Introduction to Classical and Quantum Computing* Rooted Grove, 2022
- [3] Yamaguchi, K., and Kempf, A. (2026, January 14). Encrypted qubits can be cloned. arXiv.org. <https://arxiv.org/abs/2501.02757>, American Physical Society

7.0.1 Link to Simulation

https://colab.research.google.com/drive/1YvzKDQy3GIaTiiEBT_YneUhXa4VLHe9K?usp=sharing